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MetroAT: A Benchmark Dataset for Data-Driven Maintenance in Metro Operations

Andreas Steiner^{a,*}, Omar Abdelkader^b, Stefan Helm, Fazel Ansari^a
^aResearch Unit Production and Maintenance Management, Institute of Management Science, TU Wien, 1040 Vienna, Austria

^bWIENER LINIEN GmbH & Co KG, Fahrzeugtechnik- Strategische Planung und technische Ausbildung, Erdbergstraße 202, 1030 Wien

* Corresponding author. E-mail address: andreas.steiner@tuwien.ac.at

Abstract

The reduction of breakdowns in safety-critical components of metro operations requires advanced maintenance strategies. Data-driven maintenance goes beyond traditional methods by leveraging AI techniques, but needs large datasets with operating, failure, and maintenance records. To address this need, *MetroAT* is introduced as a novel open-source dataset for metro pneumatic systems, serving as a benchmark for data-driven maintenance research in this domain. The dataset consists of 105 sensor variables and 25 million data points collected over an entire year, covering multiple components of the pneumatic system. Our contribution includes a traceable data preparation process and an event-aware operational dataset, *MetroAT*, which provides a solid foundation for AI- and ML-driven maintenance research. *MetroAT* provides sector-independent pneumatic data that support manufacturing analytics and reliability engineering for Original Equipment Manufacturers.

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1. Introduction

Urban metro systems, especially rolling stock, are critical assets that must maintain high availability and reliability to ensure safe, efficient and stable passenger transport. Operational failures in these vehicles may cause significant service disruptions, leading to increased maintenance costs and potential safety risks. To enhance operability, the industry is shifting away from traditional time-based maintenance practices towards data-driven approaches such as condition-based or predictive maintenance (PdM) [2, 3].

To enable these data-driven strategies, the effectiveness of novel analytical approaches (e.g., Machine Learning (ML)) depends heavily on the quality of the underlying data. A transparent and traceable end-to-end dataset-generation process is required, covering all stages from sensor-based data acquisition to data selection, ensuring a reproducible dataset [6]. However, implementing such data-driven methods at the asset level remains largely unexplored [5]. Consequently, most existing datasets focus on single components or isolated subsystems, offering limited support for system-level analysis and overlook-

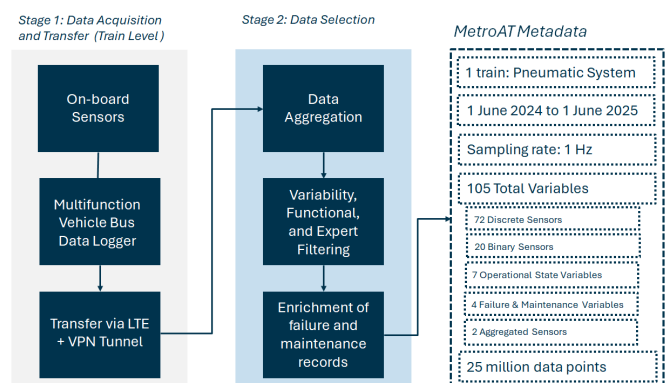


Fig. 1: Schematic representation of the *MetroAT* dataset generation process.

ing causal relationships among interacting subsystems within full-scale rolling-stock operations [7, 13]. To bridge this gap, a system-level approach is defined for one of the most critical subsystems, the pneumatic system, which accounts for approximately 20% of failures [12] and supplies compressed air for braking, suspension, and other essential functions.

This paper presents a methodology (see Figure 1) for generating an event-aware operational dataset for metro pneumatic

systems using existing sensors onboard. The contribution is twofold: (1) a system-level dataset-generation method for metro pneumatic systems, and (2) *MetroAT*, an open-source dataset that sets a new benchmark for size, variability, and transparency in metro system research, while offering transferable value for manufacturing analytics and Original Equipment Manufacturers' reliability studies.

2. Research Background

2.1. Data-driven Maintenance for Metro Trains

Data-driven maintenance in metro systems covers a wide range of subsystems, including compressors, braking circuits, suspension units and other pneumatic clients, reflecting the complexity of modern rolling stock. The air production unit (APU) has been especially studied because it's a safety-critical system and continuous monitoring enables the detection of early signs of degradation [13]. Complementary work on digital sensor show that various subsystem signals, ranging from intake valves to drying towers, support wider anomaly detection across pneumatic-system architectures [2].

Research shows that PdM for APUs faces unique challenges: failures are often not detectable by traditional threshold-based methods as the systems are exposed to varying operational loads throughout the day [11, 3]. ML techniques can overcome these limitations by learning complex patterns in multivariate sensor data, as studies have explored anomaly detection with autoencoders [11, 9], failure forecasting using LSTM models or Random Forest [9], and vector autoregression methods [1]. However, they often require large, complete and well-annotated datasets that are difficult to obtain in practice [4]. Retrofitting remains a common approach to enable monitoring but lacks scalability, as each fleet requires custom hardware integration. Moreover, most metro vehicles currently in service were not designed with built-in PdM capabilities, making systematic data acquisition an urgent challenge. Therefore, even though onboard systems are often disregarded for their limited resolution and accuracy, they provide faster integratability [14, 13].

2.2. Publicly Accessible Datasets for Metro:

Only a few open data-driven datasets for metro rolling stock maintenance exist (see Table 1). Datasets like MetroPT [13] are one of the few comprehensive datasets which monitor rolling-stock subsystems, while most public datasets for metro systems focus on operations, mobility, or infrastructure rather than monitoring rolling-stock subsystems [7]. Consequently, researchers rely only on digital sensor data [1] that may not reliably transfer to real metro environments. Furthermore, none of the existing published datasets provides a clear and reproducible process for dataset generation. A major barrier to the adoption of data-driven maintenance in metro fleets is the heterogeneity, sparsity, and inconsistent quality of the available data. Across existing datasets, data generation and selection are often incomplete, with expert knowledge rarely incorporated [8]. Gener-

ally, datasets can be broadly grouped into two categories: (i) high-dimensional datasets with many features, which capture system-level complexity but require substantial preprocessing and (ii) low-dimensional datasets with only a few monitored variables, which typically address specific components or subsystems and rarely support system-level analysis.

Table 1: Overview of publicly available pneumatic-related rolling stock maintenance datasets and the proposed *MetroAT* dataset.

Dataset	Functional Level	Sensor Specification
Metro de Madrid (Spain, 2020) [10]	Pantograph–catenary	4 variables (electrical)
MetroPT (Porto, 2022) [13]	Air Production Unit	20 variables (8 analog, 4 GPS and 8 digital)
<i>MetroAT</i> (Vienna 2025)	Pneumatic system level	105 variables (see Figure 1)

A key gap is that, to the best of the authors' knowledge, no existing dataset provides high-dimensional data with functional coverage across an entire metro train, a limitation that *MetroAT* is designed to address by offering a new benchmark dataset.

3. Methodology

The proposed methodology outlines the end-to-end process used to transform raw metro-train sensor data into an event-enriched *MetroAT* dataset. It consists of two main stages: (Stage 1) data acquisition and transfer from onboard sensors via the train internal fieldbus transmitted through a secure VPN-over-LTE connection to the backend servers, (Stage 2) data selection including data aggregation, expert feedback and functional filtering and maintenance and failure record enrichment. Throughout the paper, dataset variables are referenced in the text using the notation (*VARIABLE*).

4. Implementation Design

4.1. MetroAT Coverage

Operational Level. The *MetroAT* dataset focuses on the subsystems and components (see Figure 3b–3g) of the pneumatic system of a single metro vehicle. Structural coverage is limited to the subsystems and components of the pneumatic system, including the APU, the pneumatic brake and leveling subsystems. Data were recorded during regular passenger service on several underground lines of the metro network, with operations occurring predominantly in tunnel sections and therefore under comparatively homogeneous environmental conditions. This operational context is captured explicitly in the dataset through the variables (*TRAIN_LINE*) and (*TRAIN_CURRENT_SECTION*). The temporal coverage spans from 1 June 2024 to 1 June 2025, providing one full year of observations sampled continuously at 1 Hz. Data sources include:

- **Sensor readings (S_x):** Time-synchronised time-series data from on-board systems.
- **Failure records (F_x):** Documented service-affecting failures with disruptions longer than 3 minutes.
- **Maintenance records (M_x):** Planned revisions and unplanned post-revision repairs.

As illustrated in Figure 2, data collection runs whenever the train is powered on. Consequently, recordings are not limited to normal passenger service, but also depot movements, workshop activities and periods of degraded operation. Four partly overlapping scenarios are distinguished for the sensor data: (i) operation under normal service conditions, (ii) maintenance activities, (iii) failure periods, and (iv) intervals in which failure handling and maintenance overlap. To ensure temporal comparability across these data types, failure and maintenance events are aligned with the 1 Hz sensor time base by converting recorded event logs into discrete timestamp annotations.

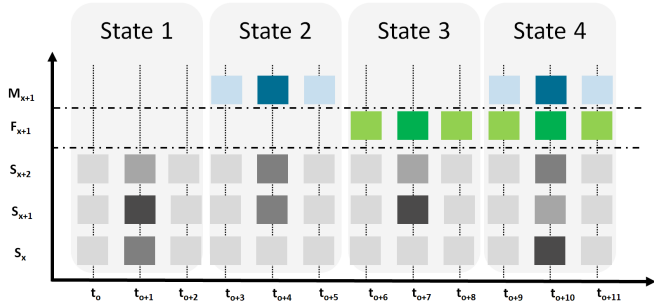


Fig. 2: Illustration of operational, maintenance, and failure states.

Train Level:. Each train is a permanently coupled six-wagon unit composed of two control wagons (CW) at the ends and four motor wagons (MW) in the middle, as shown in Figure 3a. A CW contains the driver's cabin, auxiliary converters and an APU. A MW mainly houses the traction equipment and drive components. The task of the APU is to generate a system pressure of 6.5 to 8 bar to the main reservoir line (*MAIN_RESERVOIR_PRESSURE*), which extends along the entire length of the train to supply compressed air to the most important consumer components on the vehicle, namely the pneumatic brake and leveling system. Each wagon contains two bogies with two axles each. On every bogie a pneumatic brake equipment unit (PBEU) is installed, which provides the holding brake at standstill, supplements the electrodynamic brake in case of increased braking demand, and acts as an emergency brake. Each PBEU includes a brake control computer (BCC), a pneumatic valve block, air-suspension bellows, and smaller air reservoirs. In addition, each axle has a brake cylinder and a spring-applied parking brake managed by the PBEU of the respective bogie. The proportional valve is high important, as it applies the calculated braking pressure according to the braking mode, train loading, train velocity and pressure in the main reservoir line. The *MetroAT* dataset is derived from the sensors and information connected to the PBEUs, particularly from pressures in the main reservoir, brake cylinders, proportional valves, spring brakes as well as load pressure in the air-

suspension bellows. As there are two APUs on each train, the sensor data is often redundant, allowing comparative analysis.

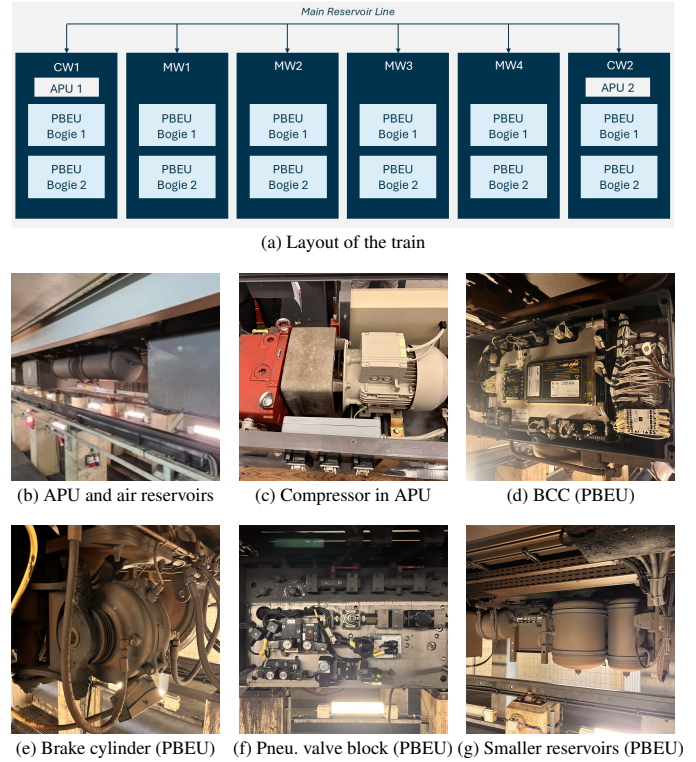


Fig. 3: (a) Schematic of the train, its wagons, and their subsystems and components. (b–g) Images of the subsystems and components.

4.2. Dataset generation process

Data Acquisition and Transfer:. The communication of the train implemented via Multifunction Vehicle Bus (MVB), defined by the IEC 61375 Train Communication Network standard. Each wagon forms a local MVB segment, and the segments are interconnected via a backbone running through the entire train. The BCCs and other pneumatic subsystems from the APUs and the PBEUs appear as MVB nodes and exchange operational data over this backbone. The metro train was retrofitted with a dedicated certificated on-board data logger solution, connected to the MVB as a read-only node. The data logger regularly polls a configured set of MVB variables, timestamps them with the train clock, and stores the resulting data stream in a compact binary format. Every ten minutes, stored files are transmitted over LTE via a secure VPN to back-end servers. An automated pipeline then decodes the binary logger format and exports the selected channels as time-stamped CSV files, forming the raw basis of the *MetroAT* dataset.

Data Selection:. The raw data streams are heterogeneous and initially distributed across numerous independent CSV exports, requiring substantial restructuring before they can be used for ML approaches. Therefore, the aim of this stage is to convert these disparate inputs into a consistent dataset through a systematic five-step approach: (1) aggregation of the data into a unified and machine-readable format, (2) identification and

grouping of variables into meaningful categories, (3) data-type-specific filtering to reduce dimensionality, (4) conducting functional and expert-driven relevance analyses to determine relevant causal sensor variables and (5) enrichment of the time series with aligned failure and maintenance records.

(1) Data Aggregation: Each train dataset consists of 120 relevant hardware ports for metro pneumatic systems, reflecting both redundant and primary subsystems. Collected data from the train is first consolidated into a unified multivariate time series via timestamp normalization and cross-file alignment. These disparate sources motivate converting the aggregated corpus into a hierarchical Apache Parquet layout, whose columnar structure and built-in compression provide superior I/O efficiency and analytical access patterns, compared with flat-file formats such as CSV or containerized stores like HDF5 [13].

(2) Data Type Clustering: In total, the raw dataset comprises 1537 unprocessed, diverse sensor variables. These are categorized as follows: (i) 679 constant sensors, which show no variation over time and primarily serve configuration or status purposes, (ii) 666 binary sensors providing on/off boolean states, such as valve positions or activation flags and (iii) 192 discrete sensors which are non binary, covering events or categorical variables such as station location

(3) Data Variability Filtering: This step is performed on the first three months of data and subsequently generalised to the full dataset. The general objective here is that sensors that exhibit a single dominant state over the observation period are excluded, as they typically correspond to static configuration parameters or inactive diagnostic channels. Out of the initial 1537 signals, 679 constant sensors are removed. For the binary sensors, a variance-based thresholding scheme is applied via a Bernoulli-distributed variable $\text{Var}(X) = p(1 - p)$, where p denotes the empirical probability of the active state. A total of 387 binary sensors are discarded because they were active or inactive for $\geq 90\%$ of the time. During window-level feature aggregation, near-constant binary channels contribute essentially no variance to the resulting features. Retaining them would inflate the dimensionality without adding informative value. This pruning step increases structural clarity and tractability, while implicitly assuming that state transitions carry the relevant information. For the 192 discrete sensors, Shannon entropy is computed using $H(X) = -\sum_{i=1}^n p(x_i) \log_2 p(x_i)$, where $p(x_i)$ denotes the empirical probability of state x_i . An entropy threshold of 0.1 is used to filter out sensors with insufficient state diversity. None of the discrete sensors falls below this threshold, so all are retained. Overall, the filtering procedure reduces the effective sensor space from 1537 to 471 signals.

(4) Functional and Expert Filtering: The functional relevance phase is essential for incorporating expert knowledge and is supported by an interactive tabular interface that documents each signal's filtering steps, constancy status, and, if applicable, the reason for exclusion. This process is iterative and continues until domain experts confirm an adequate signal set. Sensors are then filtered based on subsystem-specific functional relevance, retaining only those tied to leveling, braking, or compressor behavior. Applying this relevance filter yields 105 signals mapped onto 33 subsystem-specific sensor ports, including redundant

CW1/CW2 channels and a single aggregated train-wide measurement port.

(5) Enrichment of failure and maintenance records: In the time period, 11 Service Affecting Failures (SFA) in the pneumatic system were taken into account. These SFAs be attributed to three subsystems: 7 failures in pneumatic brake system, 3 in the APU, and 1 in the leveling system. The SFA durations range from 2 to 9 hours, with a median duration of 4.52 hours. Outliers exhibit longer durations of up to 50 hours, which may be associated with maintenance-related complications or technical issues. In addition, 18 scheduled inspection and maintenance activities were in the period. These include 9 inspection activities every four weeks, 1 maintenance activity every 24 weeks, 1 overhaul every 48 weeks, and 7 activities covering special system-level maintenance tasks. All maintenance activities are done in one day, except the yearly overhaul and HVAC-maintenance, which last 4 and 5 days. As an output of the inspection activities, 17 repair orders were placed: 9 for the pneumatic brake system, 4 for the APU, and 4 for the leveling system. These repairs address a technical problem that was solved before occurring in operation, due to successful inspection. To integrate this information into the dataset, two binary indicator columns, (*TRAIN_IS_IN_FAILURE*) and (*TRAIN_IS_IN_MAINTENANCE*), were introduced to mark whether the train is within a failure or maintenance event at a given timestamp, with the flag remaining active for the full event duration. Complementary categorical fields, (*TRAIN_FAILURE.TYPE*) and (*TRAIN_MAINTENANCE.TYPE*), record the type of failure or maintenance activity. Whenever a failure, revision or repair is active, the corresponding flag is set to one and the type field contains the associated event label; at all other times, these indicators are zero.

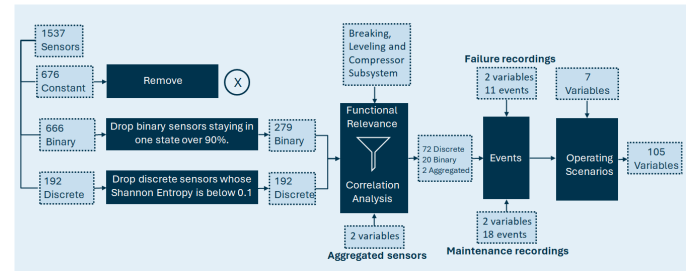


Fig. 4: Schematic representation of the detailed selection process.

4.3. Correlation Patterns of the MetroAT dataset

Sensor correlations reveal dependencies between measurements originating from physically or functionally coupled components. Strong linear associations are consistently observed across three-month windows, with many pairs showing $|r| \geq 0.99$, reflecting the train's control topology in which CW1 governs MW1/MW2 and CW2 governs MW3/MW4, resulting in synchronous actuation within each CW and consequently similar sensor trajectories (see Fig 5). Although brake-related sensors exhibit intra-group collinearity, they are not grouped because each sensor captures a distinct physical signal that may

reflect different failure modes or subsystem behaviours. Two sensor groups are fully redundant at the system level: (i) train speed (*TRAIN_SPEED_ACTUAL*), recorded independently in CW1 and CW2, but representing a single physical variable, and (2) ambient temperature (*AMBIENT_TEMPERATURE*), which samples the shared external environment. Both variables were therefore aggregated into single features. In contrast, the main reservoir line pressure exhibits lower correlations with brake-actuation pressures and forces. This behaviour aligns with its role as a supply-level variable rather than an actuator-level measurement, making it a complementary indicator of overall pneumatic system health.

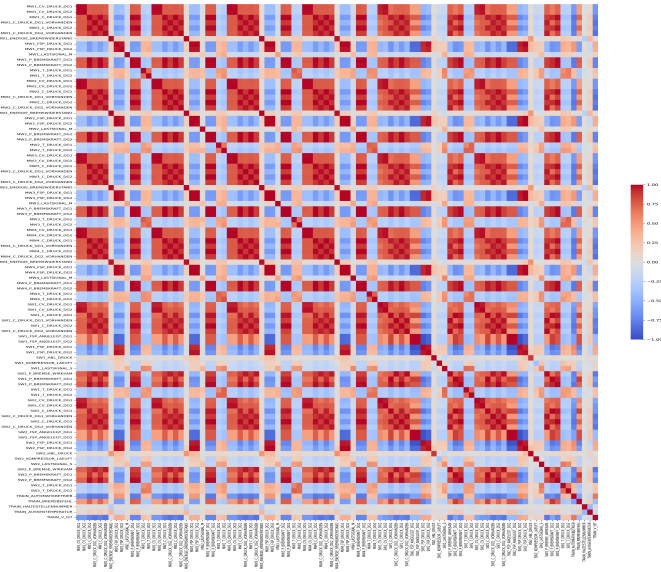


Fig. 5: Correlation matrix of selected sensors; red shows strong correlations.

Unsupervised timestamp-wise clustering (see Figure 6) was used to uncover hidden, recurring system states that emerge from patterns in the multivariate sensor data. Therefore, K-means clustering was applied by treating each high-dimensional sensor reading at a single timestep as an independent input vector, making the method operate purely spatially in feature space and fully agnostic to temporal order. After clustering, the centroids were extracted and pairwise distances between them were computed. Assuming the clusters correspond to meaningful system states, these distances quantify the dissimilarity between those states. Therefore, small clusters (e.g., Cluster 8 with 2 occurrences or Cluster 13 with 18 occurrences) located far from large clusters (e.g., Cluster 5 with 211,000 occurrences) indicate rare or abnormal operating behaviours (*i.e.*, verifiable by comparison with emergency states). Conversely, densely populated clusters suggest that normal operation consists of multiple subtly non-stationary states.

The final assembled dataset comprehensively captures both the sensor telemetry and all required contextual metadata from maintenance and failure recording, including the selected decoded signals and added status indicators. These additions ensure that every record is enriched with information about the train's real-time position and routing within the network. Key components of the *MetroAT* metadata are shown in Figure 1.

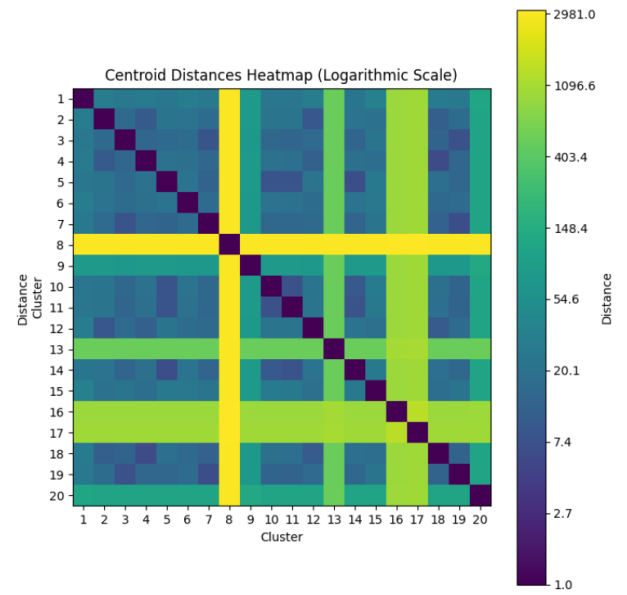


Fig. 6: Cluster centroid distance matrices using a logarithmic color scale for 20 clusters. Yellow indicates larger distances between centroids, while dark purple represents smaller distances.

5. Discussion

5.1. Potential Applications and Implications

The *MetroAT* dataset provides the first opportunity to analyze a high-dimensional set of 105 sensor signals enriched with failure and maintenance events in a metro rolling-stock context. It establishes a foundation for future subsystem-level sensor selection for targeted PdM models, structured feature creation and development of failure-specific subdatasets.

The data acquisition stage demonstrates a scalable and transferable methodology that does not require retrofitting with additional sensors. Within the data selection stage, each variable is causally assigned to a functional component, and the variance and expert-guided filtering process supports (i) ensuring capture of all relevant signals, (ii) excluding non-causal or redundant channels, and (iii) improving computational tractability through the removal of low-information sensors.

The pairwise sensor correlation and sensor-level operating states derived through timestamp-wise clustering techniques provide an initial spatial characterization of the system. The results reveal strong inter-sensor dependencies that support a sensor-group-level feature strategy, which can serve as an early layer in maintenance pipelines by flagging deviations or drifts in group behavior, even before complex models are applied. Furthermore, centroid-based clustering identifies abnormal operating states, particularly those associated with emergency-flag activations, indicating multiple operational regimes in which specific sensors exert disproportionate influence.

5.2. Limitation in the Generation Process

The removal of selected sensors inevitably reduces the available signal space and carries the risk of discarding be-

haviourally relevant information. Nevertheless, even after this reduction, the dataset remains statistically high-dimensional. Eliminating low-relevance channels improves generalization by mitigating noise amplification, making the approach well suited for broad early-warning tasks where dominant patterns prevail. However, feature pruning may suppress binary or low-frequency indicators that, despite weak global influence, can be critical in specific edge cases. The method therefore trades overall robustness and efficiency for potential losses in niche but operationally meaningful signals.

A second limitation concerns the representation of operational states. Although many states are inherently captured in the dataset, it remains unclear whether strong deviations in sensor behaviour occur across all states. Such variation may indicate that the system operates under multiple normal regimes rather than a single unified mode. At least three hierarchical layers warrant consideration: (i) local discrepancies arising from maintenance activities or operation on different track segments; (ii) the operational mode itself, such as manual operation (*TRAIN_MANUAL_MODE*), automatic operation (*TRAIN_AUTOMATIC_MODE*), or emergency mode (*TRAIN_EMERGENCY_MODE*); and (iii) procedural states, including intervals where $v = 0$ (train stationary at stations or on the track) versus dynamic states where $v \neq 0$.

6. Conclusion and Outlook

Data-driven maintenance strategies have recently gained significant traction, aiming to further reduce operational failures, particularly in metro rolling stock. These approaches, which largely build upon modern machine learning techniques, rely heavily on high-quality datasets. In this context, a comprehensive pipeline for metro pneumatic system dataset generation was proposed, encompassing the stages of data acquisition and transfer, as well as data selection and preprocessing.

Furthermore, the findings were open-sourced through the release of *MetroAT* (<https://www.kaggle.com/datasets/standreas95/metroat>), a new and more diverse dataset that addresses metro trains at the system level. *MetroAT* aims to set a new benchmark for dataset size and variability in metro systems by including 105 different sensor types along with event intervals such as maintenance and failure recordings.

Beyond metro, *MetroAT* also benefits the manufacturing community, where pneumatic systems are widely used in industrial automation, assembly, and process manufacturing. Its transferable operational and failure patterns support Original Equipment Manufacturers in data-driven service management and reliability engineering, enabling use cases such as anomaly detection, degradation modeling, and PdM.

Future work will focus on extending the temporal scope of this dataset, enhancing fleet coverage by incorporating additional assets, and improving event awareness through more detailed failure and maintenance annotations.

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